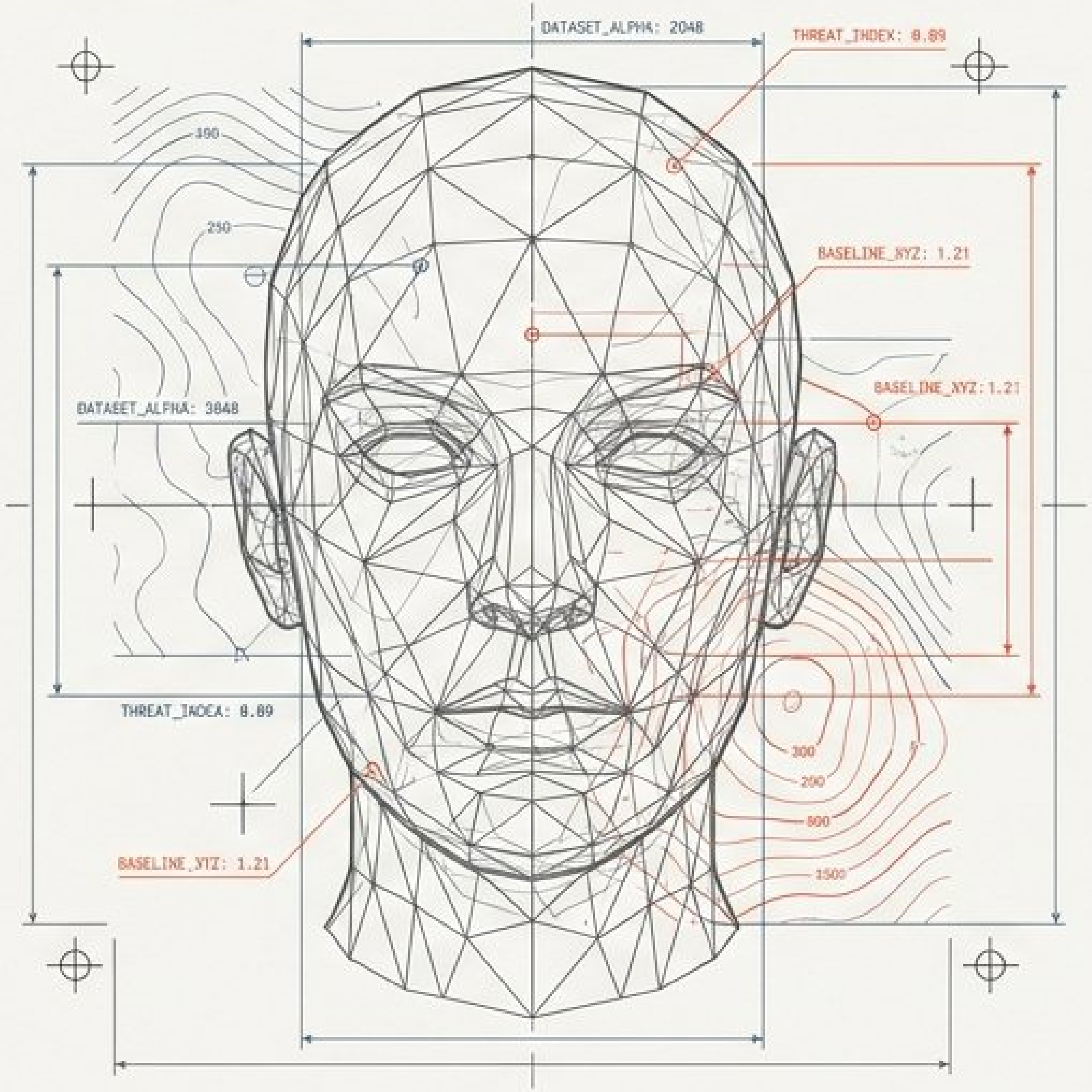
Decrypting the Synthetic Face

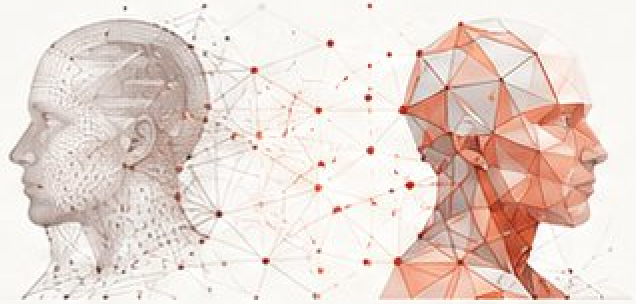
A Hybrid Deep Learning Framework for Explainable Deepfake Detection



Architecture Proposal
& Evaluation Strategy

Milestone 1: Team Contribution Tracker — Deep Learning-Based Human Face Authenticity Detection

Rohit: Foundations & Literature



Deepfake Surge

GANs and diffusion models drive realistic synthetic media.

Evolving Approaches

- Spatial CNNs
- Vision Transformers (ViT)

Challenges

- Image compression
- Unseen manipulations
- Lack of explainability

Raunak: Foundations & Literature

Benchmark Datasets

Dataset	Description	Significance
FaceForensics++	Real vs. manipulated videos	Standard benchmark
DFDC	100K+ videos, varied lighting/ compression	Real-world robustness
Celeb-DF	Real vs. manipulated videos	High-quality deepfake samples

Robust baseline evaluation strategies.

Vishakha: Analysis & Evaluation

99% Baseline Accuracy

99%

XceptionNet performs well on raw images but struggles with compression.

Hybrid vs. Single Model

Single Model → EfficientNet-B4 + ViT

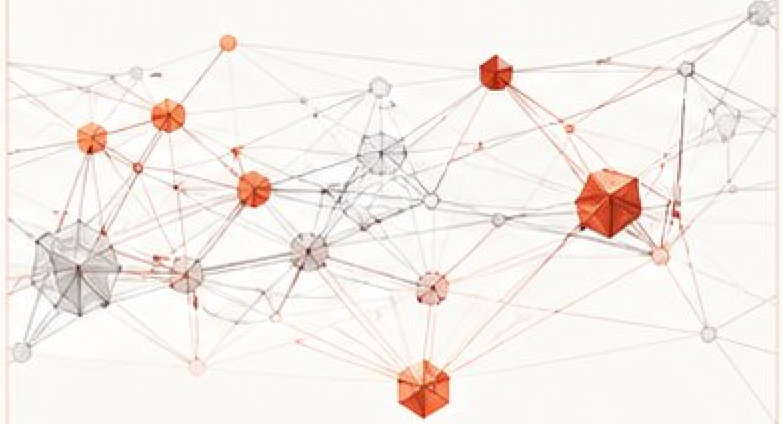
Standardized Evaluation Protocol

Using DeepfakeBench and cross-dataset testing for generalization.

Aman: Analysis & Evaluation

Key Metrics

- **ROC-AUC** — Measures classification ability
- **Grad-CAM** — Visual confidence heatmaps
- **F1-Score** — Balance of precision /recall



Robust & explainable deepfake detection.

The Synthetic Media Threat is Scaling Exponentially

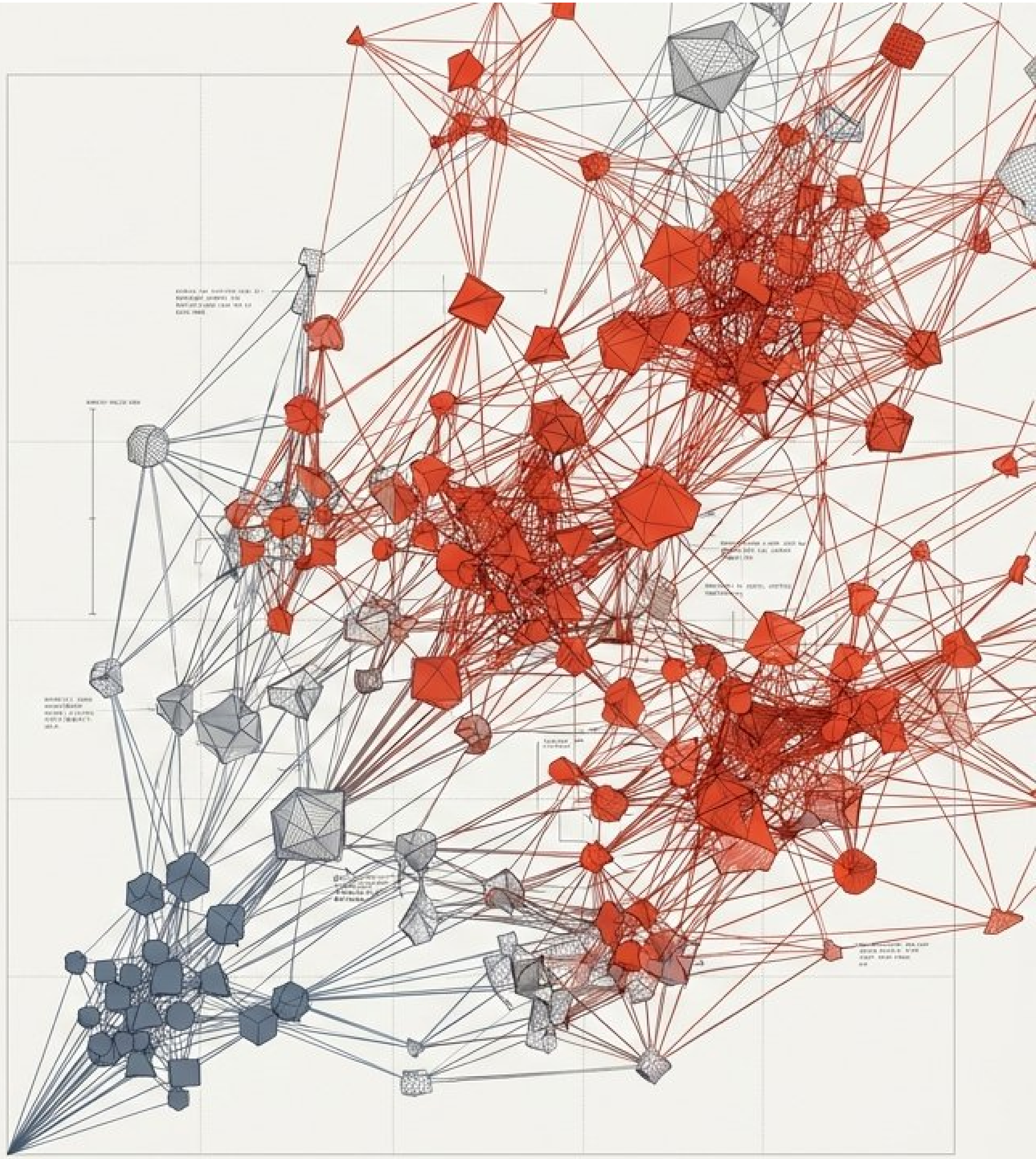
Generative AI, driven by advanced GANs and diffusion models, has commoditized the creation of hyper-realistic facial media. This technology is now routinely exploited for misinformation, identity theft, and digital fraud.

ANNUAL_DATASET_GROWTH: +300%

Misinformation: Undermining trusted digital communication channels.

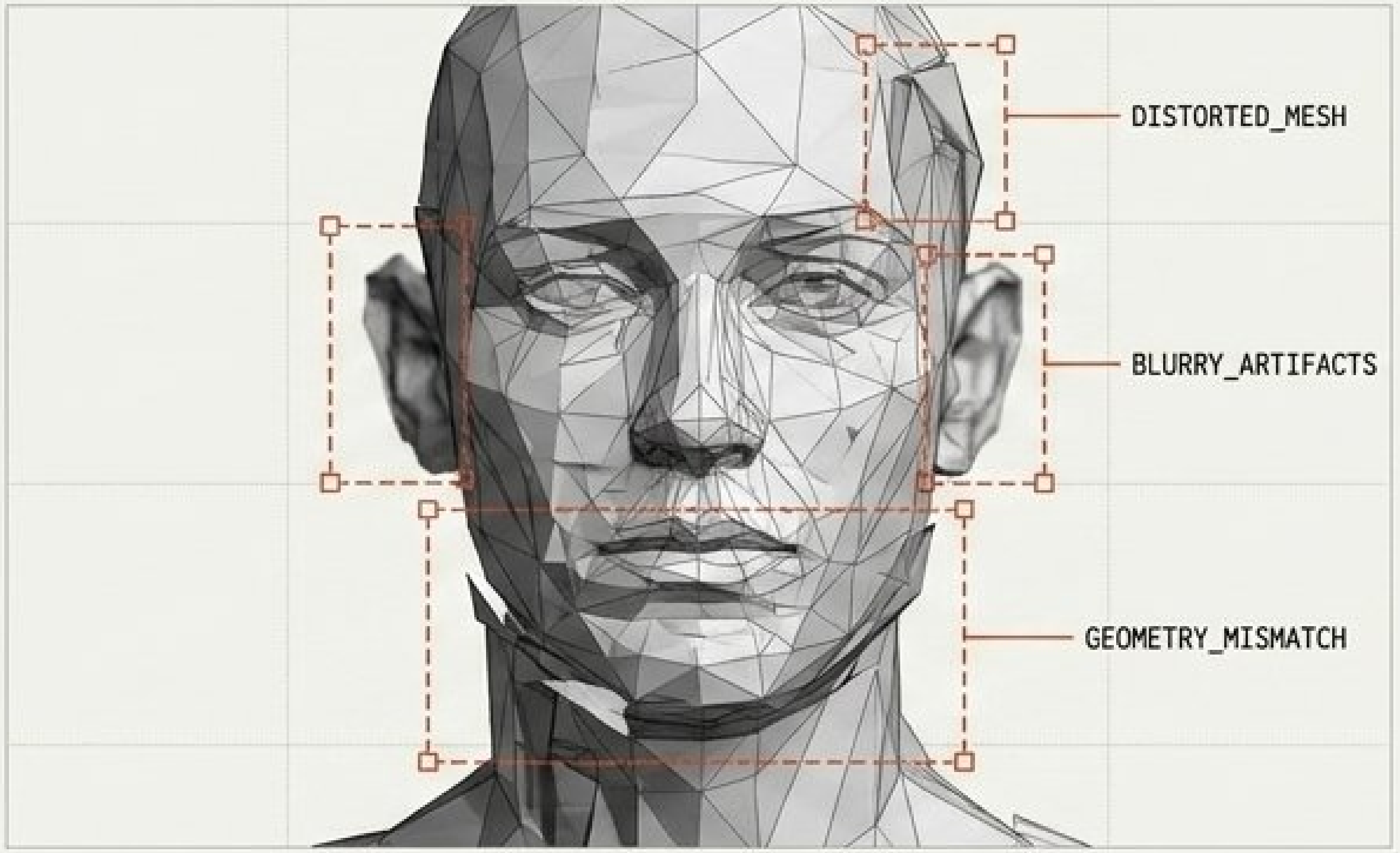
Impersonation: Bypassing biometric security and identity verification.

Digital Fraud: Enabling sophisticated social engineering attacks at scale.



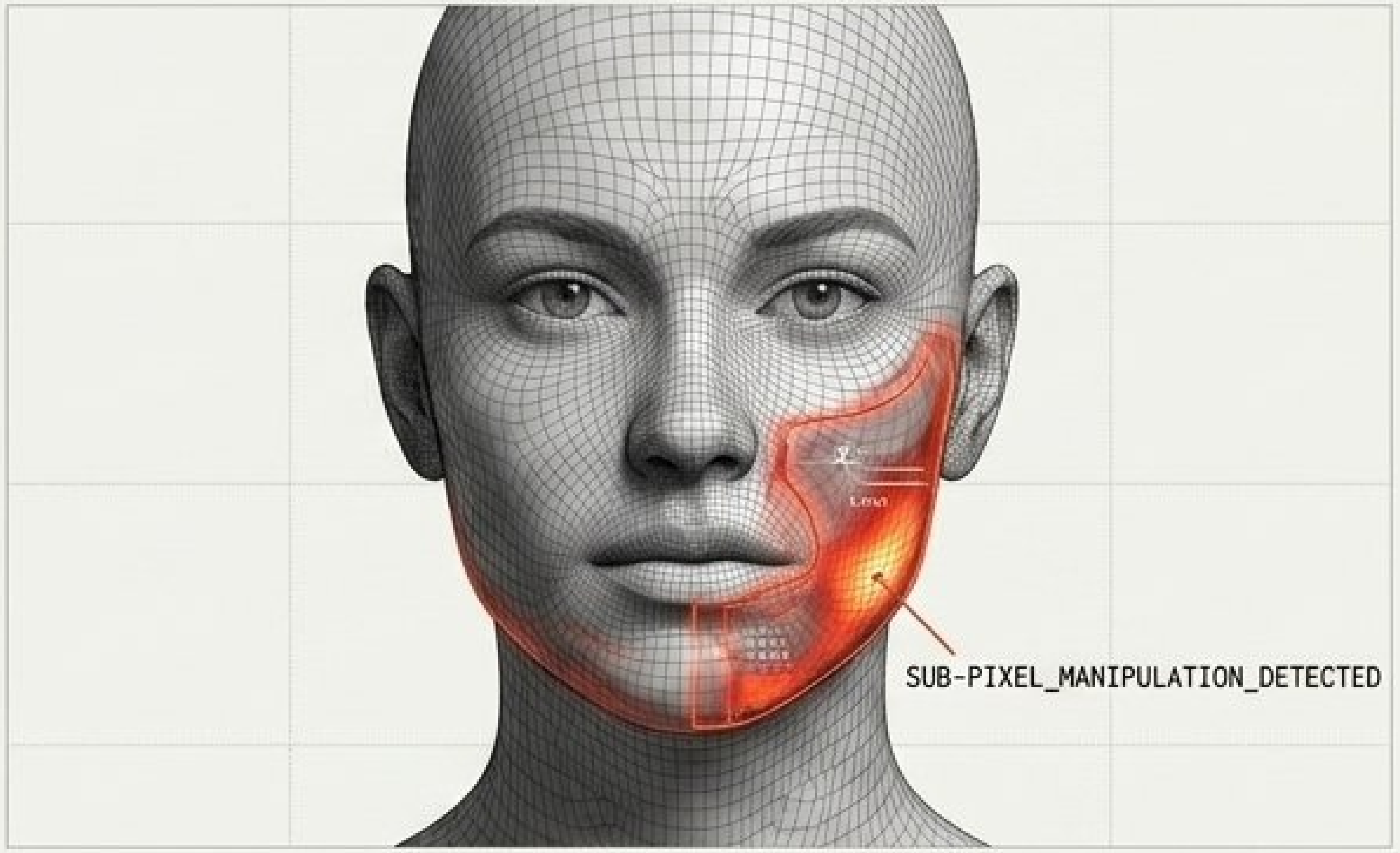
The Rapid Evolution of Forgery Artifacts

Early Generation - ≈2018



- Facial Details: **Blurry ears, mismatched eyes, distorted teeth**.
- Geometry: **Incorrect shadows, hair artifacts, inconsistent backgrounds**.
- Detection: Easily intercepted using conventional **CNNs**.

State-of-the-Art - Current



- Facial Details: **Realistic skin texture, consistent features**.
- Geometry: **Accurate facial geometry**, **natural lighting** and **reflections**.
- Detection: Requires advanced **deep learning** to detect **sub-pixel artifacts**.

Existing Detection Mechanisms and Their Limitations

Architecture	Target Features	Strengths	Limitations
Spatial CNNs (e.g., XceptionNet)	Skin texture and lighting.	Fast inference, 99% accuracy on benchmarks.	Overfits to specific datasets; poor cross-dataset generalization.
Vision Transformers (ViT)	Global facial structure and symmetry.	Better cross-domain generalization.	High computational cost; requires massive training datasets.
Frequency/Wavelet	Invisible spectral artifacts.	Catches what spatial models miss.	Highly sensitive to image compression.
Noise Residual	Camera sensor fingerprints.	Isolates synthetic vs. hardware signatures.	Unreliable after heavy compression or editing.
Multi-Scale	Global and local artifacts.	Highly robust.	Extreme computational complexity.

The Evaluation Battlefield: Standard Benchmark Datasets

DeepFake Detection Challenge (DFDC)

`TOTAL_CLIPS: 119,197`

Varying compression levels, lighting, and backgrounds. The ultimate measure for real-world robustness and cross-domain generalization.

FaceForensics++ (FF++)

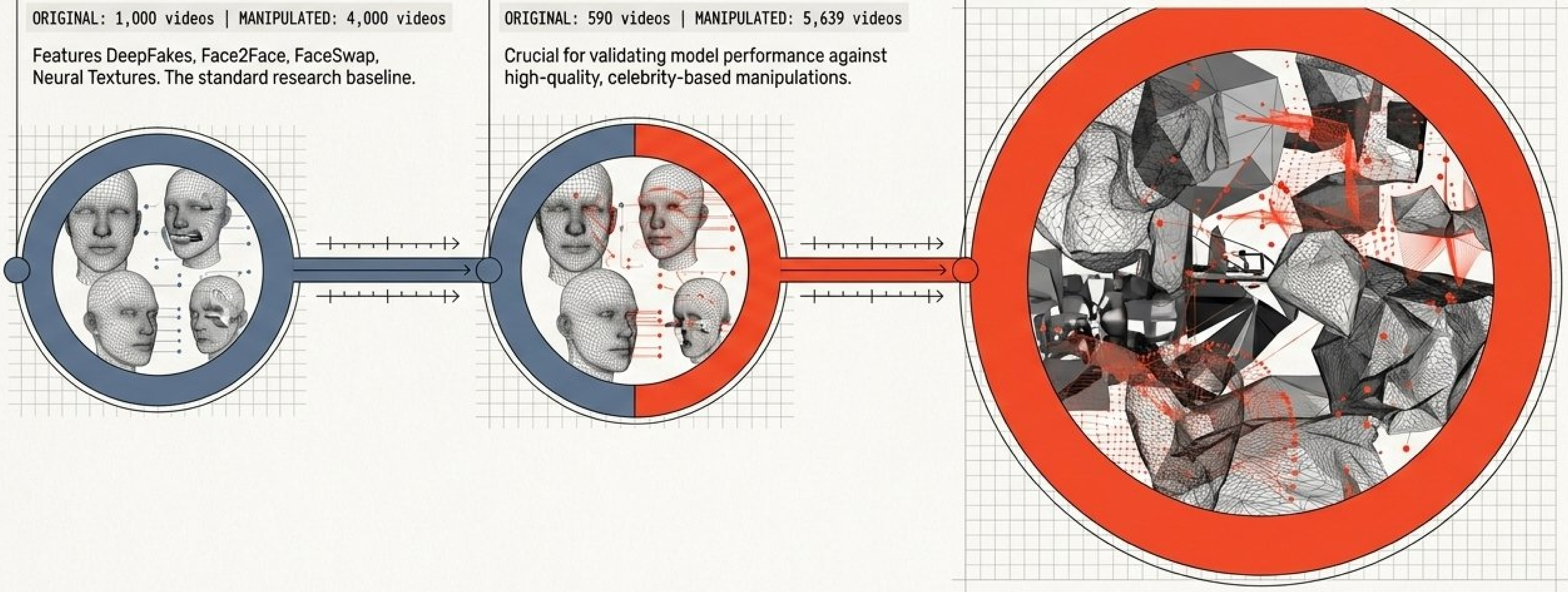
`ORIGINAL: 1,000 videos | MANIPULATED: 4,000 videos`

Features DeepFakes, Face2Face, FaceSwap, Neural Textures. The standard research baseline.

Celeb-DF v2

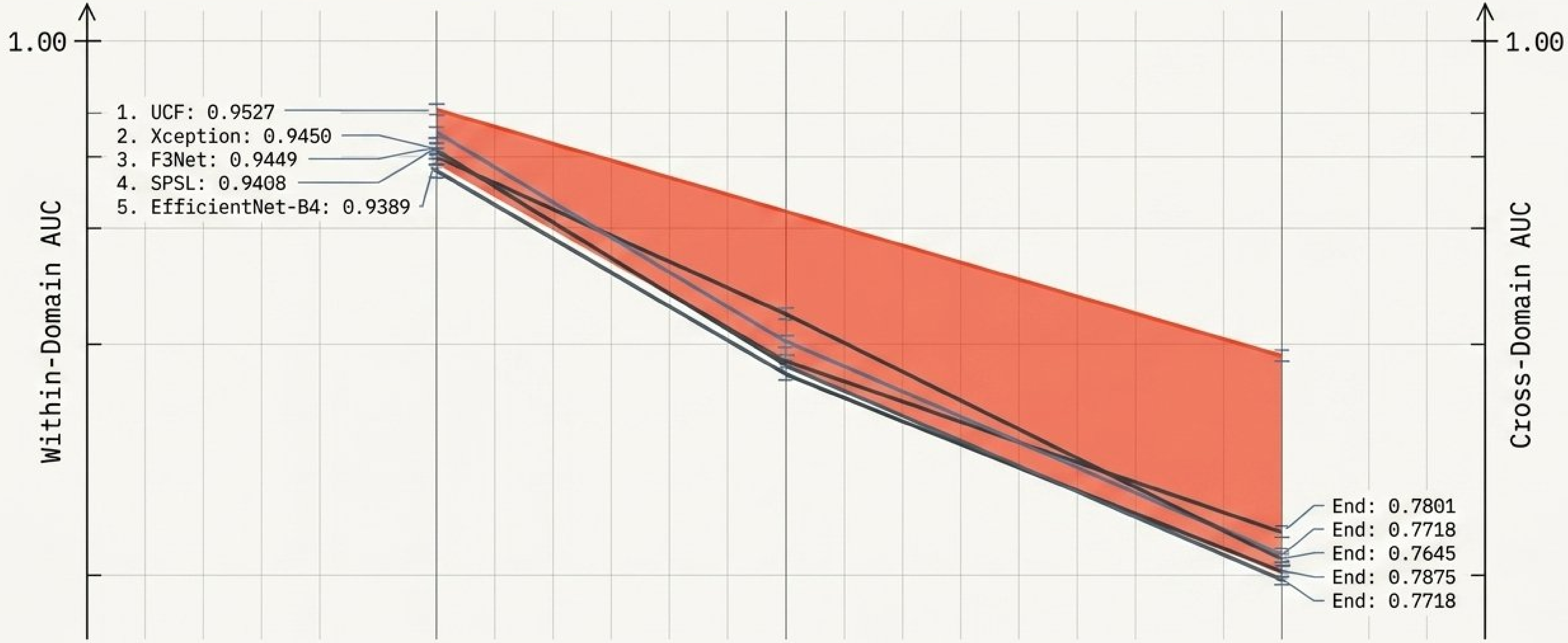
`ORIGINAL: 590 videos | MANIPULATED: 5,639 videos`

Crucial for validating model performance against high-quality, celebrity-based manipulations.



The Generalization Drop: Why Current Baselines Fail

High benchmark accuracy is an illusion.
Models overfit to their training distribution.
When tested against unseen datasets (cross-domain), detection capabilities collapse.

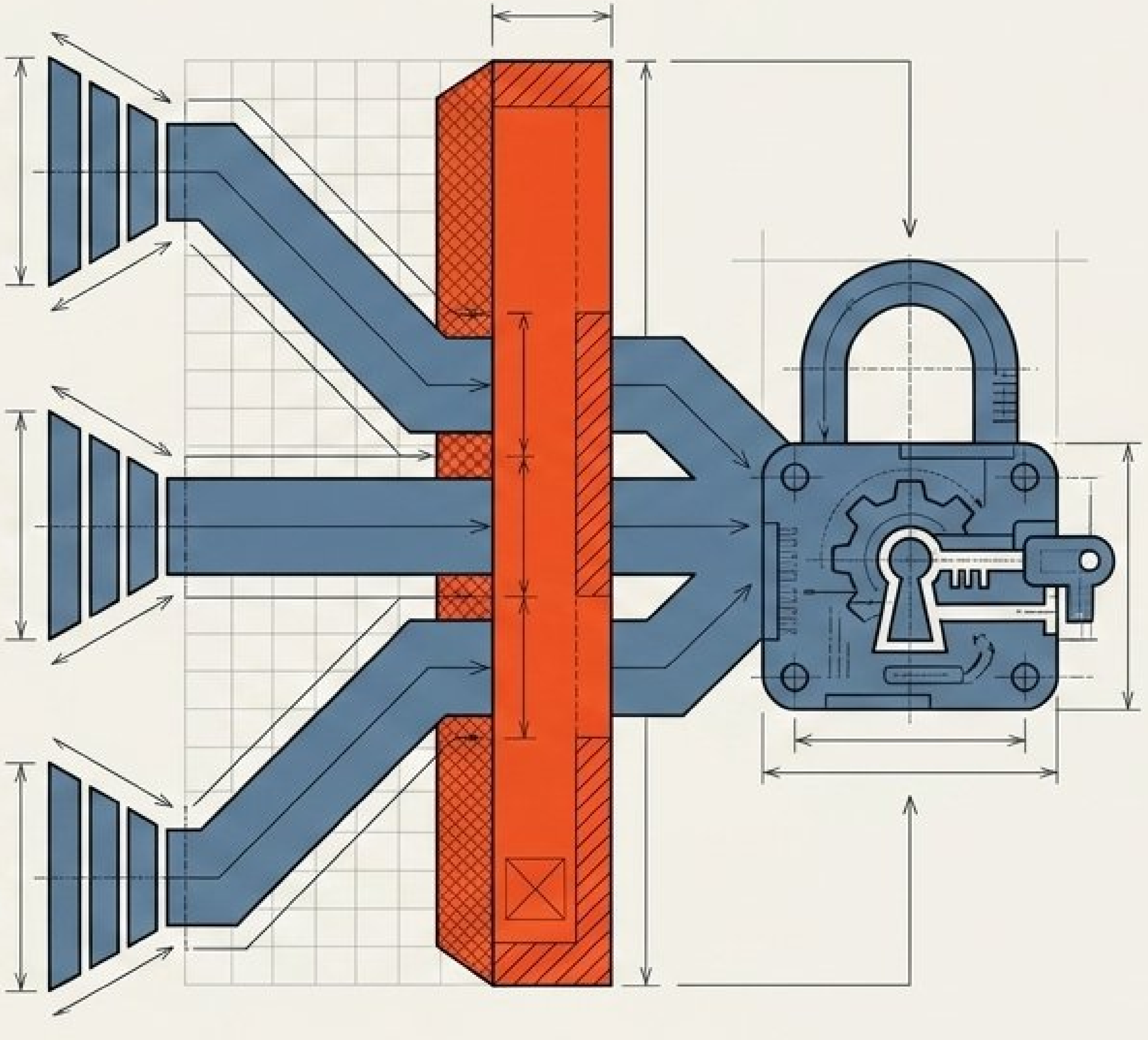


The Research Gap: The Necessity of a Unified Approach

The Feature Gap:
CNNs memorize local textures but miss global context. ViTs see the whole face but miss micro-artifacts.

The Compression Vulnerability:
Real-world media is compressed (WhatsApp, X, YouTube), stripping the high-frequency forensic artifacts existing models rely on.

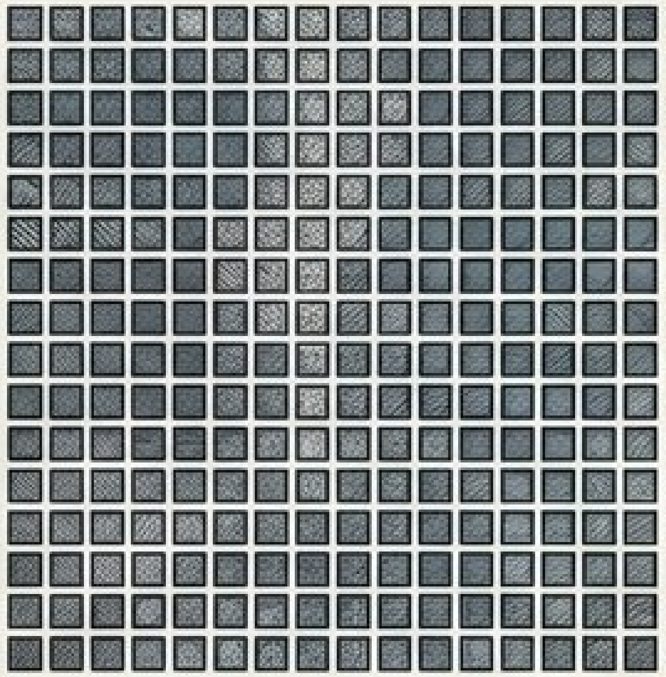
The Trust Deficit:
Current models output a binary fake/real score with zero explanation.



No single-method architecture can survive real-world, compressed, cross-domain deepfakes. True robustness requires a hybrid feature-fusion system paired with explainable AI.

The Solution: A Hybrid Deep Learning Framework

Core Concept: We propose a novel architecture that intentionally fuses the spatial sensitivity of Convolutional Neural Networks with the global contextual awareness of Vision Transformers, explicitly engineered for cross-dataset generalization.



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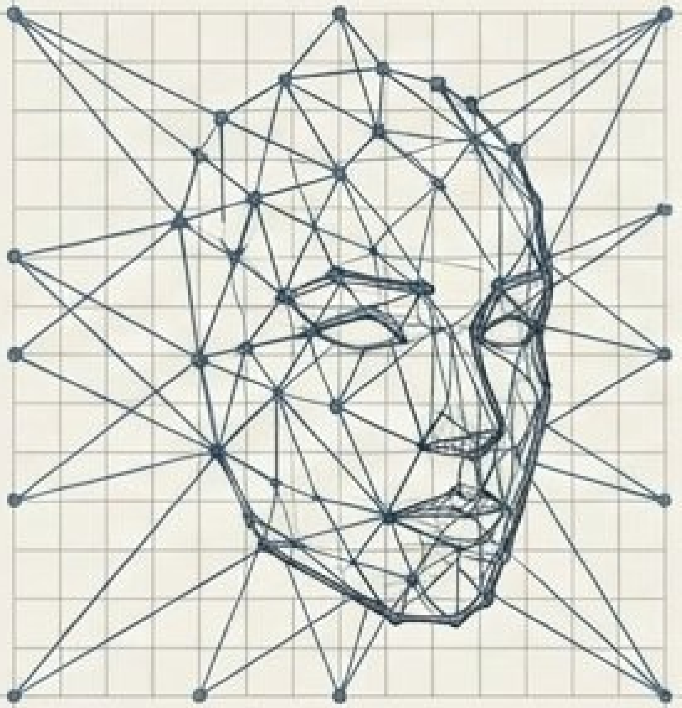

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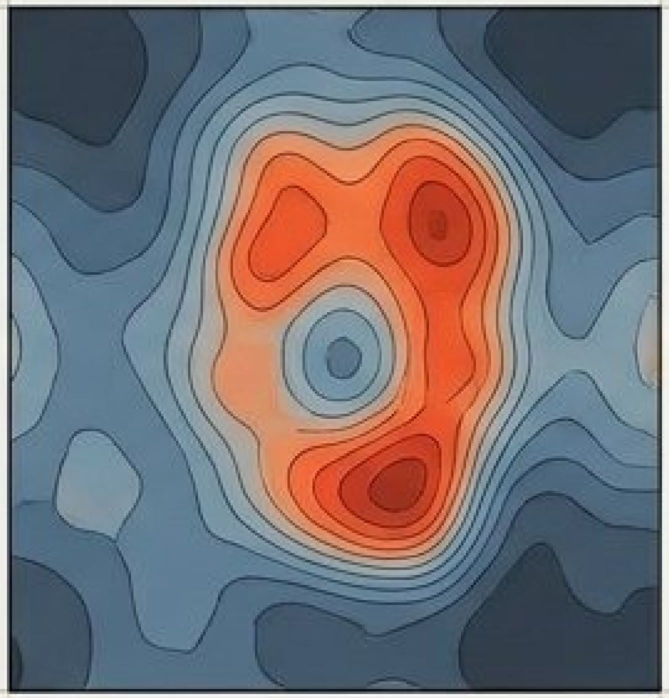

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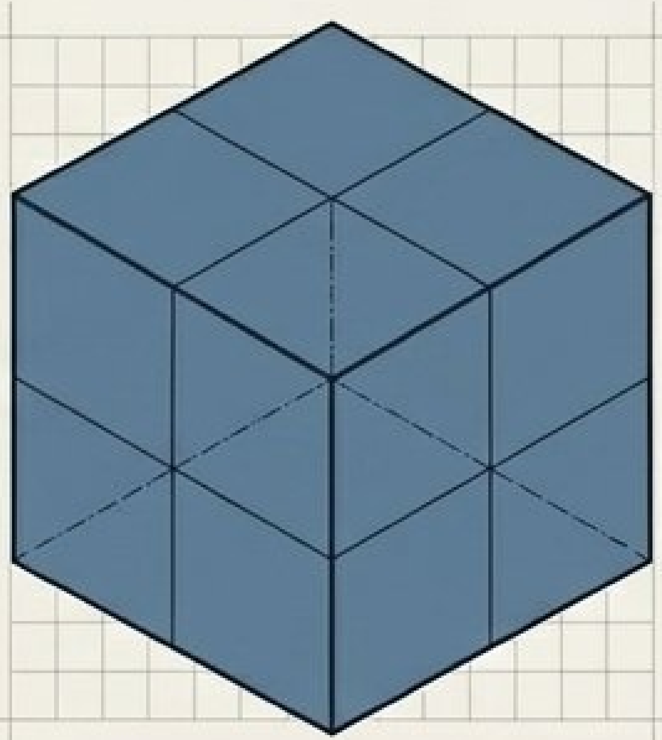


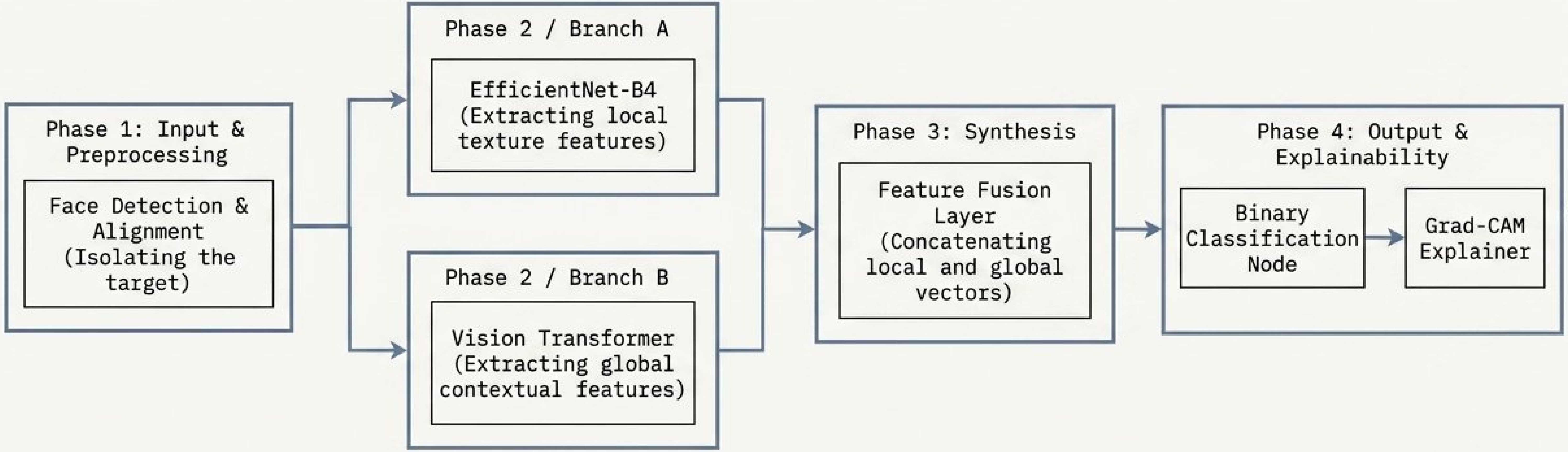
Pillar 1: Local Mastery.
Leveraging EfficientNet-B4 for highly efficient spatial and texture artifact extraction.

Pillar 2: Global Context.
Utilizing ViTs to map long-range dependencies, symmetry, and illumination consistency.

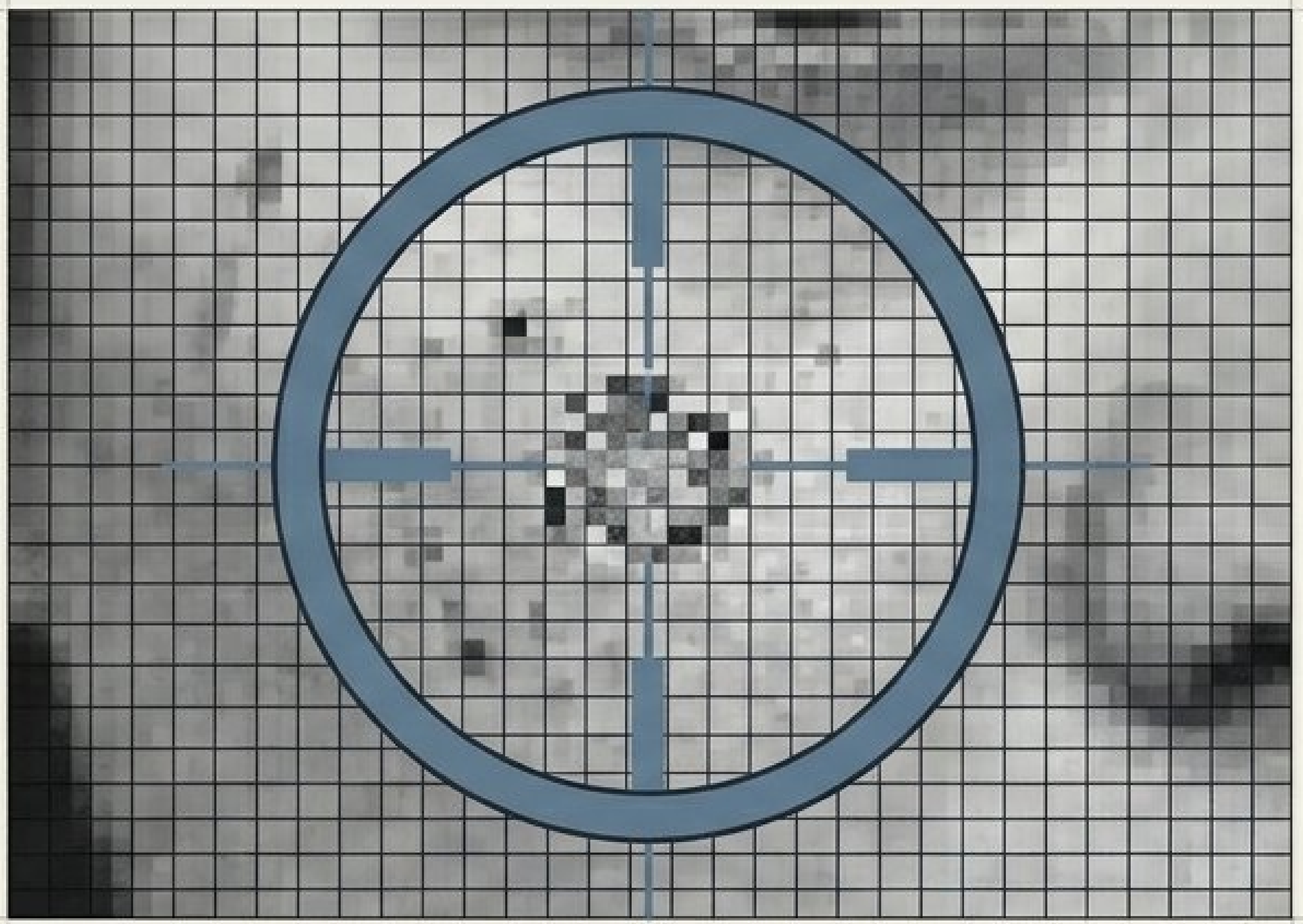
Pillar 3: Transparency.
Integrating Grad-CAM to render confidence heatmaps, transforming a black-box classification into a glass-box forensic tool.

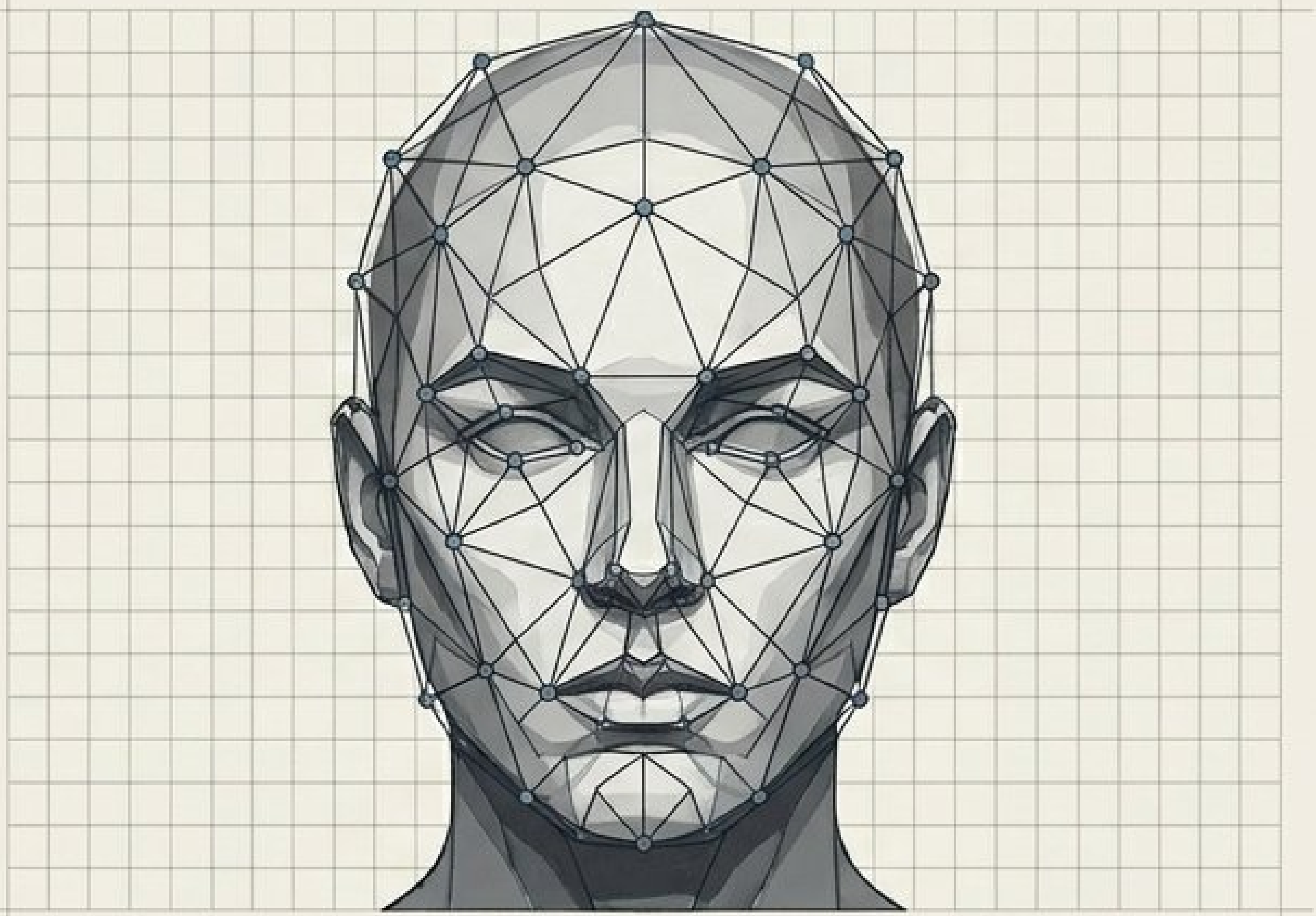
The Proposed Framework.

Architectural Blueprint



Feature Fusion: Bridging the Micro and Macro





Local Features via EfficientNet-B4

The Micro: Analyzes pixel-level inconsistencies, blending boundary errors, and localized synthetic noise introduced by GANs.

Advantage: Highly parameter-efficient extraction of spatial artifacts.

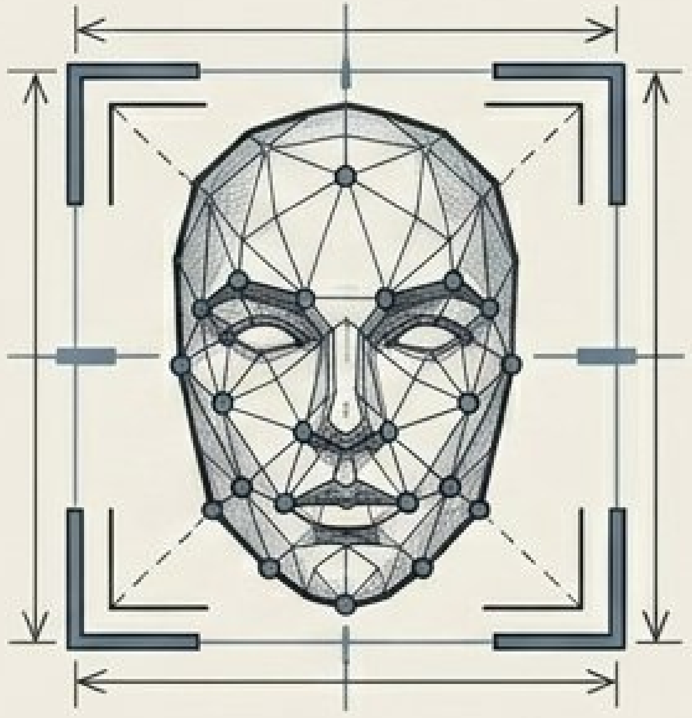
Global Context via ViT

The Macro: Divides the image into patches to learn long-range dependencies. It understands if the lighting on the left cheek matches the right, or if the facial symmetry is mathematically coherent.

Advantage: Overcomes the CNN tendency to overfit to localized dataset quirks.

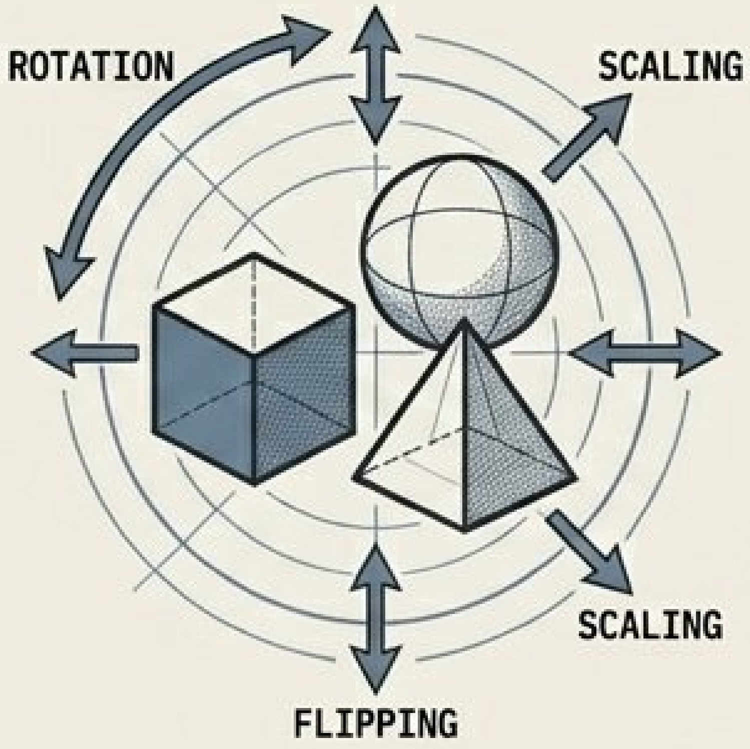
Engineering for Real-World Robustness

Engineering for Real-World Robustness



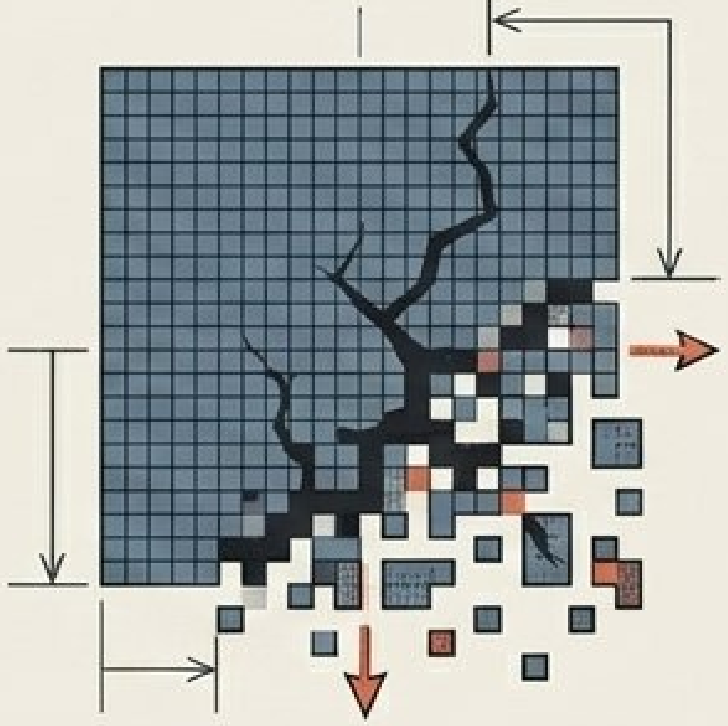
Face Detection & Alignment

Raw media contains irrelevant background noise. Preprocessing strictly isolates and aligns facial landmarks to ensure the model only evaluates relevant biometric geometry.



Heavy Augmentation

The training pipeline heavily utilizes rotation, flipping, and brightness adjustments to prevent the model from memorizing specific environmental conditions.



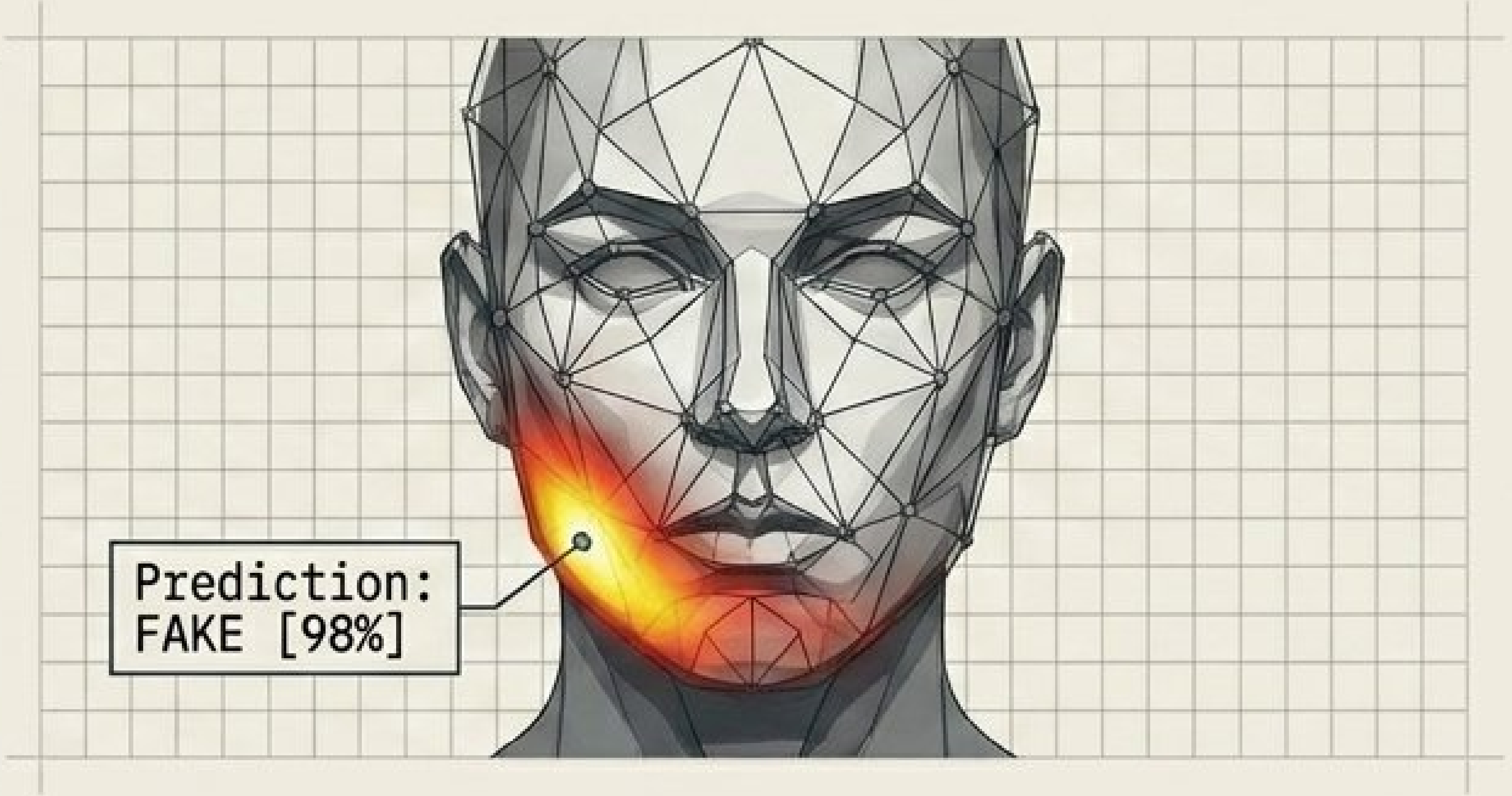
Compression Simulation

The Critical Step: Because social platforms strip data, the model is intentionally trained on dynamically compressed data. It learns to detect the structural signatures of deepfakes even when high-frequency artifacts are destroyed.

Explainability as a Core Feature

A high-accuracy system is useless if human analysts cannot trust its output.

Black Box vs. Glass Box



Commercial tools provide raw classification scores without indicating which regions triggered the prediction.

The Glass Box - Grad-CAM Integration
Gradient-weighted Class Activation Mapping (Grad-CAM) generates confidence heatmaps. It highlights the exact spatial regions (e.g., mismatched blending boundaries) driving the model's decision, restoring user trust and transparency.

The Paradigm Shift: Defining the Innovation

Current Standards	**Proposed Framework**
Isolated CNNs or Vision Transformers.	Hybrid EfficientNet-B4 + ViT with active Feature Fusion.
Local OR Global bias.	Simultaneous local texture and global context extraction.
Minimal (raw frame extraction).	Rigorous Face Detection & Alignment.
Optimizes for within-domain benchmark scores.	Engineered specifically for cross-dataset survival.
Unexplainable black-box outputs.	Visual Grad-CAM heatmaps for analyst verification.

Evaluation Strategy & Target Metrics

TRAIN
FaceForensics++ (Learning foundational artifacts).

→

VALIDATE
Celeb-DF (Tuning against high-quality manipulation).

→

TEST
DeepFake Detection Challenge [DFDC] (The ultimate cross-domain test).

Baseline Competitors: Xception, EfficientNet-B4, UCF, F3Net, SPSL.

[ROC-AUC] - Primary metric for classification threshold robustness.

[ACCURACY]
Compact metric value.

[PRECISION]
Compacted precision.

[RECALL]
Complioned recall.

[F1-SCORE]
Compacted F1- value.

[CONFUSION MATRIX] - For mapping false positive/negative distribution.

Expected Outcomes & Real-World Impact

Outcome 1: Cross-Domain Dominance

Achieve statistically significant improvements in cross-dataset generalization compared to standalone CNN or ViT baselines.

Outcome 2: Compression Resilience

Deliver consistent ROC-AUC performance even on low-quality, heavily compressed social media distributions.

Outcome 3: Forensic Scalability

Provide a scalable, highly transparent framework suitable for immediate deployment in identity verification, media forensics, and digital fraud prevention.

Moving deepfake detection from theoretical benchmark optimization to reliable, real-world human face authenticity verification.